Learning SysML Modeling for Space Applications

Team M102 – Europa Submarine Model Research: Communication Subsystem

AE 4699 MAV Undergraduate Research
Academic Credit: 2 hour

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1. Team M102 - Model Purpose & Overview

The team for this project consisted of the following undergrad students: Dhriti Naik, Anish Kulkarni, Ryland Marley, Aaditya Patel, Toby Reitman, and William Mulder.

The goal of this project was to model the various systems of a theoretical NASA mission to Europa using a SysML model in MagicDraw. The basis for this mission takes inspiration from NASA's Clipper mission and SWIM prototypes. Our theoretical mission would see a spacecraft travel to Europa, land, and drill into its subsurface oceans to then deploy the SWIM submarine probes.

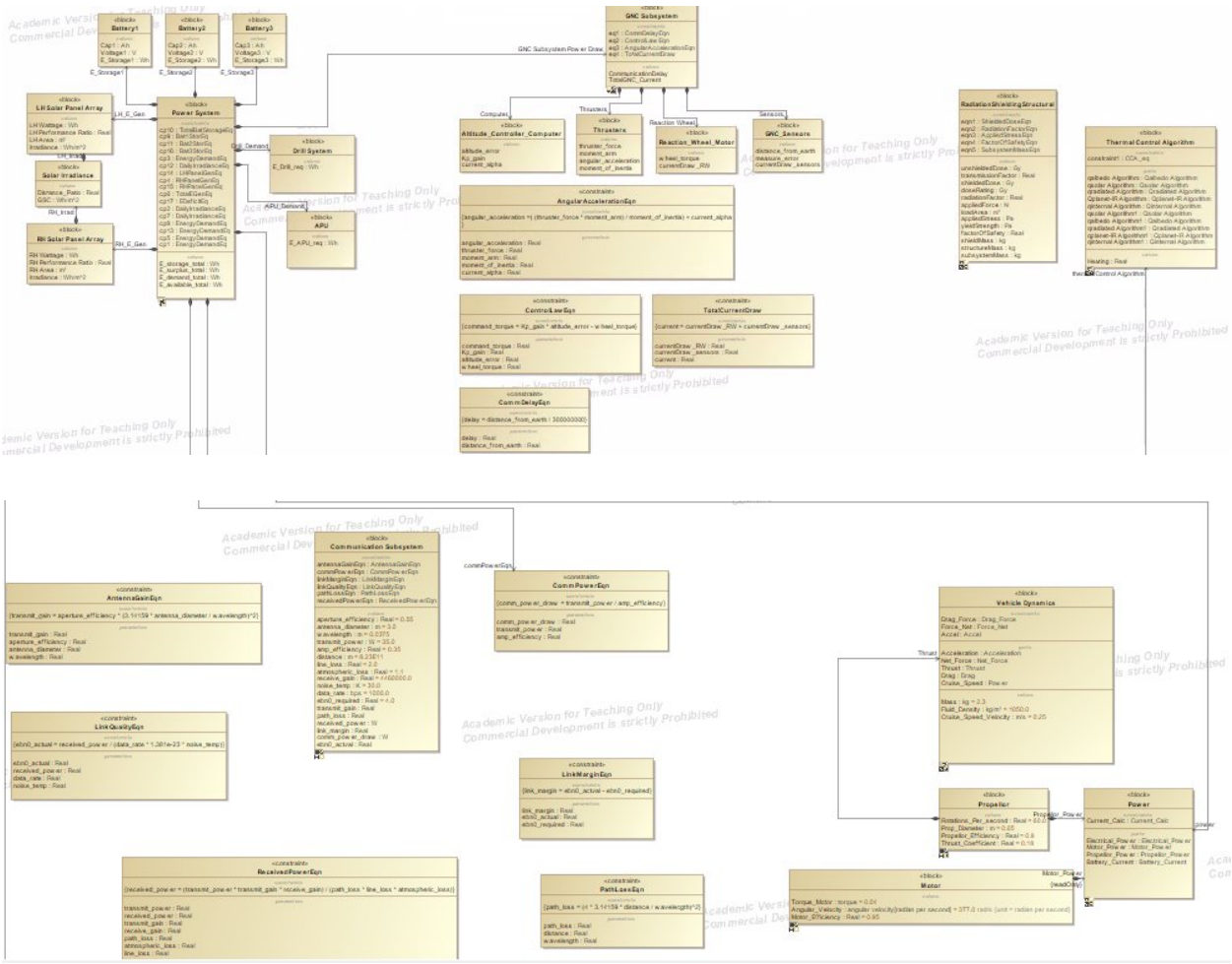


Figure 1 – Europa Submarine Model, whole-system block definition diagram (bdd) showing all six subsystems and their component blocks.

The model is composed of six subsystems: Thermal Control, Power, Radiation Shielding and Structural, Communications, Propulsion, and GNC (Guidance, Navigation, and Control). Within each subsystem, we have implemented relevant equations that allow us to simulate values that are important to the spacecraft on its mission using MagicDraw's simulation toolkit. These equations are represented through parametric diagrams, while block definition diagrams help to show how the elements of each subsystem relate to each other. The subsystems are all connected to each other

through an overall power equation, which calculates how much power needs to be drawn from the batteries. Together, these subsystems model the spacecraft and submarine, while calculating mission critical values to give a representation of this theoretical mission.

2. Overview of My Work – Anish Kulkarni

Purpose of the Communication Subsystem

The Communication subsystem attempts to solve one engineering problem: will the science data transmitted from Europa be decodable at Earth with acceptable quality, and how much electrical power would this consume? The distance between Earth and Europa varies between 4.2 to 6.5 AU, at which point the signal received at Earth would be attenuated to 28 orders of magnitude below the emitted value. The objective of the model is to prove that in spite of the attenuation, the combination of antenna diameter, transmitting power, and data rate will ensure sufficient energy and bit to decode the signal at the Deep Space Network.

The Communication subsystem requirements are outlined below:

- Ensure the radio frequency downlink from the Europa surface lander to Earth at 5 AU.
- Prove that there is enough signal received to decode the science data.
- Provide information on the power consumption by the radio to the Power subsystem.

Model Architecture

The subsystem was represented by the single Communication Subsystem block, containing all the value properties, with six constraint blocks defining the governing equations. The value properties were assigned the proper value types wherever possible: transmit power, received power and communication subsystem power draw were assigned the type W; the values for antenna diameter, wavelength and distance were assigned the type m; noise temperature was assigned the type K; and the data rate was assigned the type bps. The ratios of gain, loss, efficiency and energy per bit are dimensionless ratios and therefore correctly defined as the value type Real.

Values were stored on the main subsystem block and not split between components because of the need for the parametric wiring to be straight and for each equation in the chain to access the same set of properties. Distance, data rate and communication subsystem power draw were the only properties that crossed the subsystem boundaries and were therefore assigned at the highest level of the block in order to preserve external bindings.

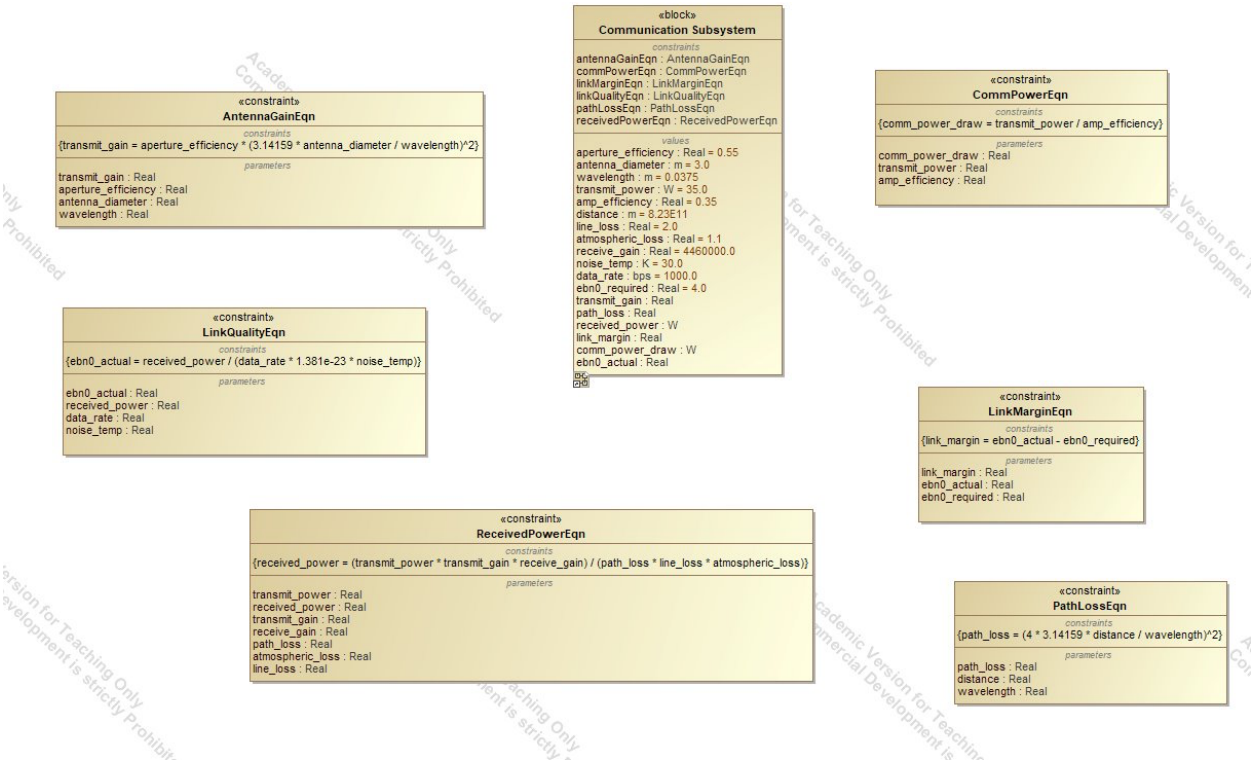


Figure 2 – Communication Subsystem block definition diagram (bdd), showing the main block and its six constraint blocks.

Governing Equations

Six mathematical expressions make up the subsystem. The first five make up one solve chain that depicts the route taken by the signal from the transmitter to the ground station, while the sixth depicts the cost of electricity required for transmission.

Equation	**Expression**	**Purpose**
Antenna Gain	$G = \eta\,(\pi D / \lambda)^2$	Concentration of transmitted power into a directed beam by the spacecraft dish.
Free-Space Path Loss	$Lfs = (4\pi d / \lambda)^2$	Attenuation of the signal as it spreads spherically over the Europa to Earth range.
Received Power	$Pr = (Pt \cdot Gt \cdot Gr) / (Lfs \cdot Ll \cdot La)$	Signal power arriving at the Deep Space Network antenna.
Link Quality	$Eb/N0 = Pr / (R \cdot k \cdot Ts)$	Energy per bit relative to the receiver noise floor; the true measure of decodability.
Link Margin	$M = (Eb/N0)actual - (Eb/N0)required$	Verification metric. A positive value means the link closes.
Comm Power Draw	$Pcomm = Pt / \eta pa$	Electrical power drawn from the bus, accounting for amplifier inefficiency.

The variables are defined as follows:

- η - aperture efficiency of the spacecraft antenna (dimensionless)
- D - antenna diameter (m); λ - wavelength, where $\lambda = c/f$ (m)
- d - range from Europa to Earth (m)
- Pt - radiated transmit power (W); Gt, Gr - spacecraft and ground station antenna gain (dimensionless)
- Ll - combined line and antenna pointing loss; La - atmospheric loss at the receiver (dimensionless)
- R - data rate (bps); k - Boltzmann constant, 1.381×10^{-23} J/K; Ts - system noise temperature (K)
- ηpa - power amplifier efficiency (dimensionless)

Flow of calculations is also sequential. Together, the values of antenna gain and path loss determine the value of received power. Together, the values of received power and data rate determine the value of ratio of energy per bit to noise density. This ratio, as compared to the minimum required by the receiver and its coding scheme, yields the link margin as the sole pass or fail criterion for the subsystem. Equation for comm power consumption also uses the value of transmit power as an input and provides an output of the subsystem.

Gain Gt is that of 3 m antenna mounted on spacecraft and is determined in the model. Gain Gr is that of 34 m antenna of the Deep Space Network on Earth and is a known parameter, being a property of existing ground infrastructure equipment. Both parameters go into the numerator of received power equation, since improvement in focus on both ends of the link is equal.

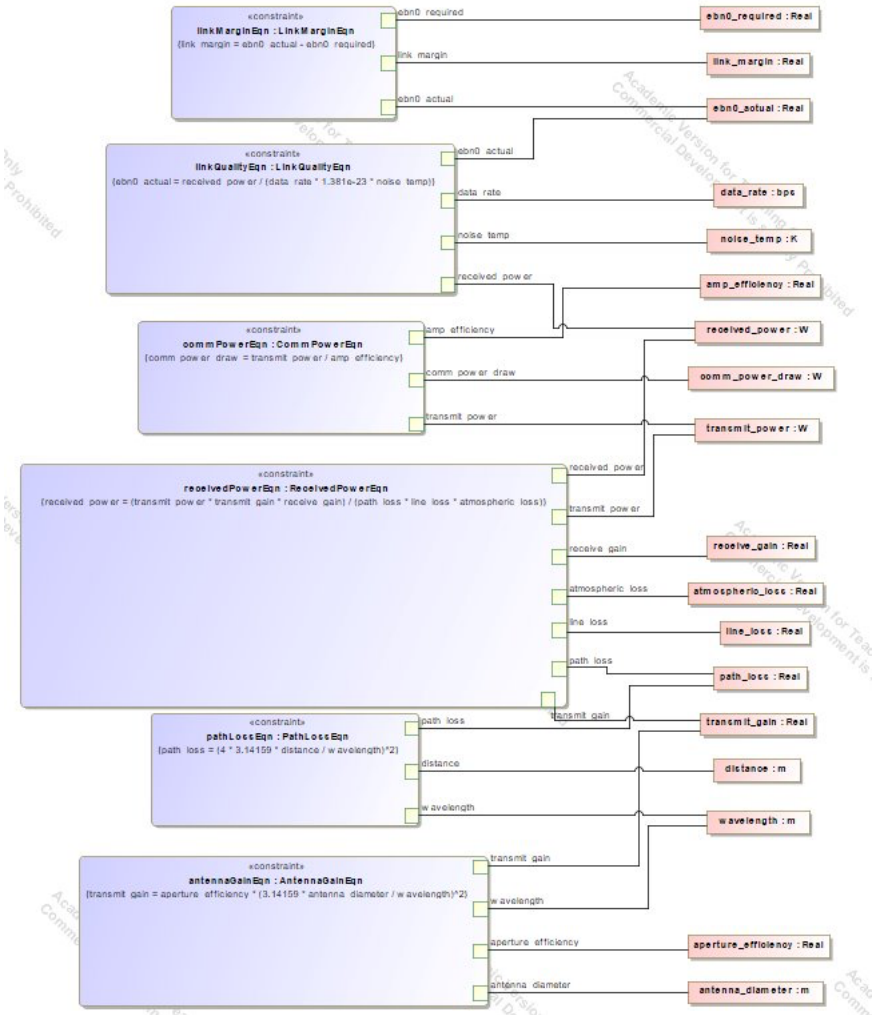


Figure 3 - Communication Subsystem parametric diagram (par), showing the six constraint blocks bound to the block's value properties.

Input Values and Their Justification

These input values were chosen to correlate with actual mission hardware parameters as opposed to being arbitrary values so as to ensure relevance of the derived margin. The antenna diameter of 3.0 m is comparable with the high gain antenna used by the Europa Clipper, where NASA has missions to Europa, and whose diameter is about 3 m. The wavelength of 0.0375 m translates to X band with a frequency of 8 GHz, which is the deep space downlink frequency for Clipper. The transmit power of 35 W represents the power used by the traveling wave tube amplifiers aboard the Clipper. The receive gain and the system noise temperature represent the 34 m Deep Space Network station equipped with a cryogenically cooled low noise amplifier.

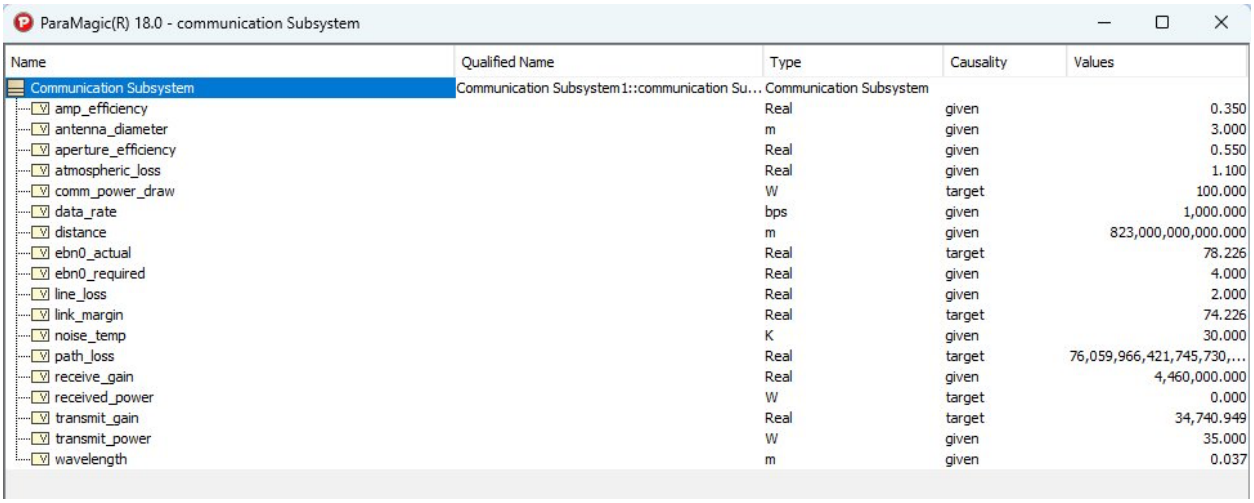
Value property	**Type**	**Value**	**Basis**
aperture_efficiency	Real	0.55	Typical reflector antenna efficiency
antenna_diameter	m	3.0	Europa Clipper high-gain antenna
Wavelength	m	0.0375	X-band downlink, ~8 GHz

Value property	Type	Value	Basis
transmit_power	W	35.0	Representative deep-space amplifier output
amp_efficiency	Real	0.35	Typical traveling wave tube amplifier efficiency
distance	m	8.23×10^{11}	Representative Europa-to-Earth range (5.5 AU)
line_loss	Real	2.0	Combined line and pointing loss
atmospheric_loss	Real	1.1	Atmospheric attenuation at the ground station
receive_gain	Real	4.46×10^{6}	34 m DSN antenna at X-band
noise_temp	K	30.0	Cryogenically cooled DSN receiver
data_rate	bps	1000	Assumed science downlink rate
ebn0_required	Real	4.0	Minimum ratio for reliable decoding

The data rate continues to be an assumption instead of an input from some other subsystem. As there is no instrument or data handling subsystem modelled by the team, no one is the owner of this input.

Instance and Solved Results

An instance of the Communication Subsystem block was created with all value properties and all six constraint properties included, and solved using the ParaMagic plugin with the twelve inputs above set as given and the remaining properties set as targets.



Name	Qualified Name	Type	Causality	Values
Communication Subsystem	Communication Subsystem1::communication Su...	Communication Subsystem		
amp_efficiency		Real	given	0.350
antenna_diameter		m	given	3.000
aperture_efficiency		Real	given	0.550
atmospheric_loss		Real	given	1.100
comm_power_draw		W	target	100.000
data_rate		bps	given	1,000.000
distance		m	given	823,000,000,000.000
ebn0_actual		Real	target	78.226
ebn0_required		Real	given	4.000
line_loss		Real	given	2.000
link_margin		Real	target	74.226
noise_temp		K	given	30.000
path_loss		Real	target	76,059,966,421,745,730,...
receive_gain		Real	given	4,460,000.000
received_power		W	target	0.000
transmit_gain		Real	target	34,740.949
transmit_power		W	given	35.000
wavelength		m	given	0.037

Figure 4 -ParaMagic browse window showing the solved Communication Subsystem instance, with given inputs and computed targets.

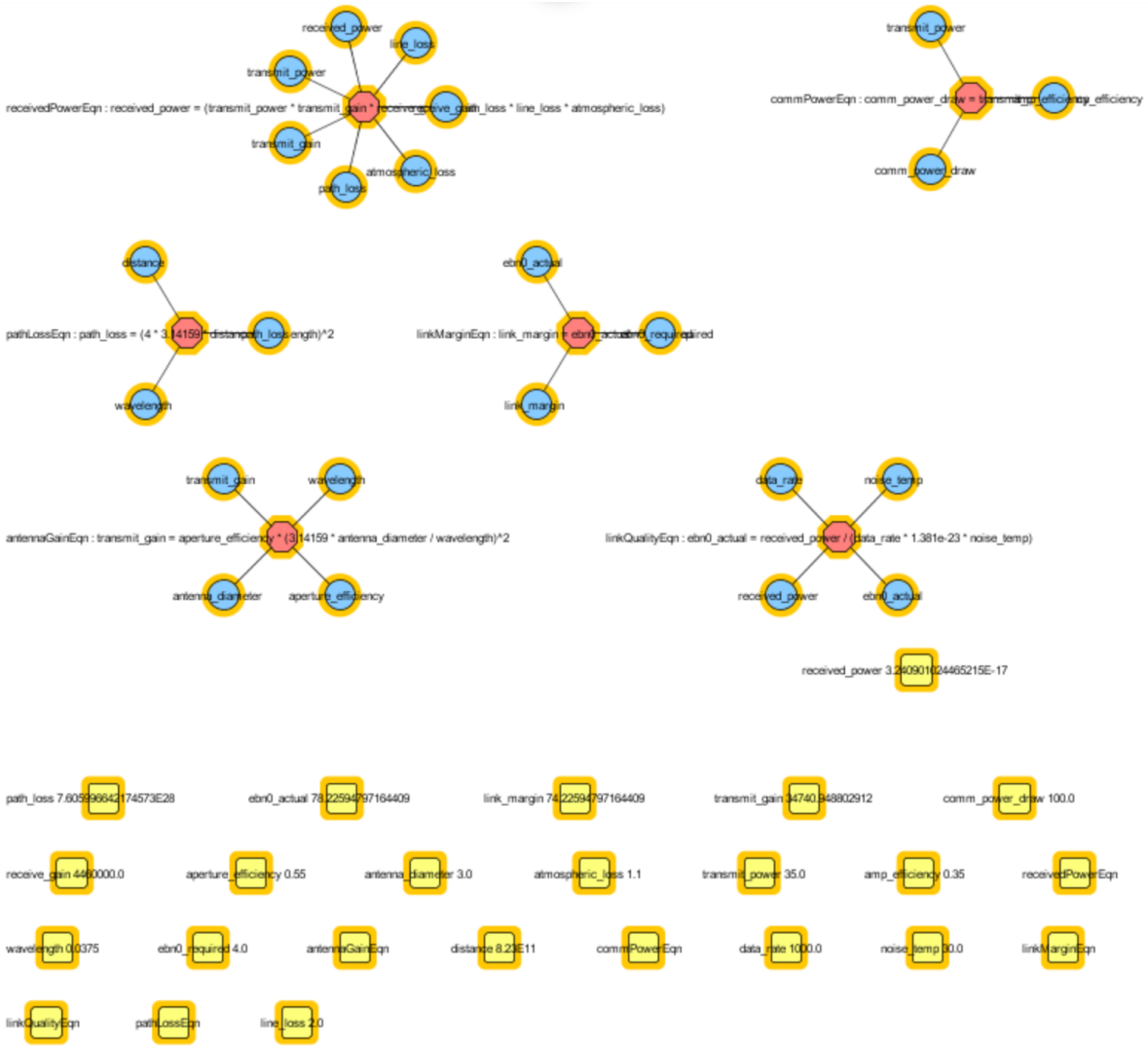


Figure 5 - BuzzToys Panorama for Communication Subsystem.

The model solved the full chain and returned the following results:

Computed property	**Result**	**Interpretation**
transmit_gain	3.474×10^{4}	45.4 dB of gain from the 3 m dish
path_loss	7.606×10^{28}	286.4 dB of free-space loss over 5.5 AU
received_power	3.24×10^{-17} W	Approximately 32 attowatts arriving at the ground station
ebn0_actual	78.23	18.9 dB of energy per bit to noise density
link_margin	74.23	Positive; approximately 12.9 dB of margin

Computed property	Result	Interpretation
comm_power_draw	100.0 W	Electrical draw reported to the Power subsystem

The link margin is positive and this is the key outcome of the subsystem: assuming the above hardware, the downlink will close with margin to spare. The magnitudes are also physically reasonable. The received power of about 3×10^{-17} W is very small indeed, but the small data rate of 1000 bps and the small system noise temperature of 30 K ensure that the energy per bit will be more than sufficient to decode. This demonstrates the design compromise at the heart of deep space communication: since it is impossible to increase the received power, the data rate is decreased to the point where each bit has sufficient energy.
The comm power consumption of 100 W derives from radiating 35 W with an amplifier that has 35% efficiency. The other 65 W is wasted as heat. This is why the same number is significant to the Thermal Control subsystem.

3. References

1. Jet Propulsion Laboratory. “DSN Telecommunications Link Design Handbook, 810-005.” NASA Deep Space Network. https://deepspace.jpl.nasa.gov/dsndocs/810-005/

2. NASA Science. “Standing Tall: Europa Clipper's High-Gain Antenna.” NASA. https://science.nasa.gov/missions/europa-clipper/europa-clipper-resources/standing-tall-europa-clippers-high-gain-antenna/

3. NASA Langley Research Center. “Europa Clipper High-Gain Antenna Undergoes Precision Testing at NASA Langley.” NASA. https://www.nasa.gov/centers-and-facilities/langley/europa-clipper-high-gain-antenna-undergoes-precision-testing-at-nasa-langley/

4. Mazarico, E., et al. “The Europa Clipper Gravity and Radio Science Investigation.” Space Science Reviews, 2023. https://link.springer.com/article/10.1007/s11214-023-00972-0

5. Bhaskaran, S., et al. “Telecommunications Systems for the NASA Europa Missions.” IEEE Aerospace Conference, 2017. https://ieeexplore.ieee.org/abstract/document/8058576/

6. NASA Science. “Europa Clipper Mission Overview.” NASA. https://science.nasa.gov/mission/europa-clipper/

7. NASA Jet Propulsion Laboratory. “SWIM: Sensing With Independent Micro-swimmers.” NASA Innovative Advanced Concepts.